Cadmiel Avenado

QUESTION 1: ESSAY

(a) According to most correlations, what are the two **variables** that can change external mass transfer coefficient?
(b) How can the mass transfer resistance be reduced?

Given Answer:

a) The mass transfer coefficient is defined as diffusivity over the distance of the boundary layer and the actual catalyst. SOme correlations that can help determine MT coefficient are the Sh number and Sc number and Re number would help determine Sh empirically. However, Kc is also proportional to V/d where V is velocity of the fluid and d is particle diameter.

b) KC resistnace = 1/Kc can be reduced by simply increasing the value of Kc. You can increase Velocity, and maybe decrease particle diameter.

Score: 25

QUESTION 2: ESSAY

1.

In a heterogeneous catalytic reaction, the measured order of reaction was different from the actual one. In fact, the $n_{actual} = 2.n_{measured} – 1$. Explain why the actual reaction-order was different.

Given Answer:

Falsified Kinetics! here Eta assumes every T inside the catalyst is same as T outside catalyst. In exothermic reactions, for example, Tinside > Toutside.

Score: 20

QUESTION 3: ESSAY

1.

For a reaction A → B on a catalyst surface, the process is mass-transfer limited. The concentration in the bulk is C_{Ab} and on the surface is C_{AS}. The rate of surface reaction is given by $r_s = k_s C_{AS}^2/(1+K_A C_{AS})^2$. If the mass-transfer coefficient is kc, what is the overall rate of reaction?

Given Answer: if the rate of reaction is mass transfer limited, then the overall reaction will be given by the reaction rate of MT because during the reaction, all rates should be equal. And since MT is limiting factor, then K''>> Kc and then -rA'' =Kc Cab

Score: 20

QUESTION 4: ESSAY

1.

An irreversible, reaction-limited, dual-site reaction A + B → C + D is taking place on a catalyst surface. When the P_D is doubled, the rate of reaction stays the same, while when P_C was doubled, the rate reduced. If, A and B both adsorb on the catalyst surface, write the most likely reaction rate equation.

Given Answer: Dual site adsorption. where KaPa +KbPb and also KcPc are present in the denominator. and because it's adsorption a 1+ at the bottom. sooo something like ---- > -ra' = KPa/(1+KaPa+KbPb)

Score: 10